

Leveraging the Latent Diffusion Models for Offline Facial Multiple Appropriate Reactions Generation

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ABSTRACT

Offline Multiple Appropriate Facial Reaction Generation (OMAFRG) aims to predict the reaction of different listeners given a speaker, which is useful in the scenario of human-computer interaction and social media analysis. In recent years, the Offline Facial Reactions Generation (OFRG) task has been explored in different ways. However, most studies only focus on the deterministic reaction of the listeners. The research of the non-deterministic (i.e. OMAFRG) always lacks of sufficient attention and the results are far from satisfactory. Compared with the deterministic OFRG tasks, the OMAFRG task is closer to the true circumstance but corresponds to higher difficulty for its requirement of modeling stochasticity and context. In this paper, we propose a new model named FRDiff to tackle this issue. Our model is developed based on the diffusion model architecture with some modification to enhance its ability of aggregating the context features. And the inherent property of stochasticity in diffusion model enables our model to generate multiple reactions. We conduct experiments on the datasets provided by the ACM Multimedia REACT2023 and obtain the second place on the board, which demonstrates the effectiveness of our method.

CCS CONCEPTS

• **Computing methodologies** → **Temporal reasoning**; • **Human-centered computing** → *Systems and tools for interaction design*; • **Applied computing** → Law, social and behavioral sciences.

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MM '23, October 29–November 3, 2023, Ottawa, ON, Canada

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ACM ISBN 979-8-4007-0108-5/23/10...\$15.00
<https://doi.org/10.1145/3581783.3612865>

KEYWORDS

Listener Reaction, Offline Reaction Generation, Diffusion Model

ACM Reference Format:

Jun Yu, Ji Zhao, Guochen Xie, Fengxin Chen, Ye Yu, Liang Peng, Minglei Li, and Zonghong Dai. 2023. Leveraging the Latent Diffusion Models for Offline Facial Multiple Appropriate Reactions Generation. In *Proceedings of the 31st ACM International Conference on Multimedia (MM '23)*, October 29–November 3, 2023, Ottawa, ON, Canada. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3581783.3612865>

1 INTRODUCTION

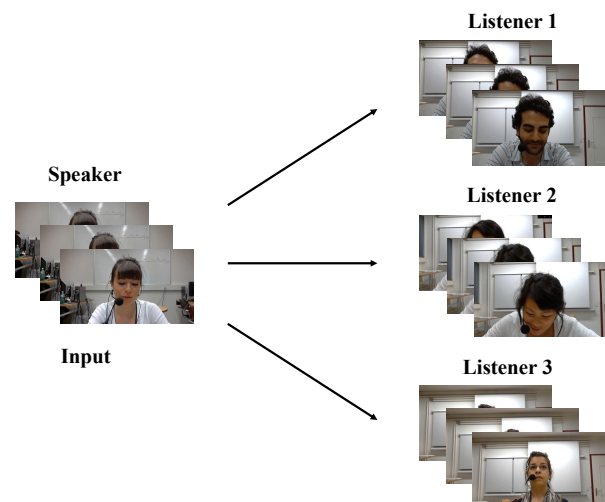


Figure 1: Multiple appropriate reaction generation in dyadic interaction settings.

In recent years, as the human-computer interaction and social media analysis develop, Offline Facial Reactions Generation

(OFRG), as an essential problem in these fields, attracts more and more attention. Generally, the OFRG task aims to predict the reaction of the listener given a speaker (see Fig.1). Numerous studies [2, 3, 7, 8, 12, 16, 17, 19] have already explored the issue before and proposed effective solutions. However, most of the studies focus on the deterministic OFRG task and neglect the fact that different people respond differently after receiving the same stimuli of the surroundings, and the same person may have a variety of appropriate response when facing the same scene because of their diverse internal state. However, a recent study [11] has been conducted to generate multiple listener reactions based on a speaker's behavior due to its higher requirement in learning stochasticity and context.

To address the challenges, we introduce the Facial Reaction Diffusion (FRDiff) Model inspired by the recent work of Barquero et al. [1]. For a better model the distribution of latent variables for facial expressions of both the speaker and listener, we integrate 1D convolutions into the Latent Diffusion Models (LDM) [15, 20]. Subsequently, FRDiff leverages the improved LDM for the stochastic generation of facial reactions.

To summarize, our main contributions are as follows:

- We propose the FRDiff model based on LDM to predict the facial reaction of the listener. In comparison to previous methods, our approach strikes a balance between diversity and appropriateness, and significantly outperforms two baseline models in terms of diversity metrics.
- We integrate 1D convolutions into the LDM and design the CPM module to better model the distribution of latent variables for facial expressions of both the speakers and listeners.
- We conduct experiments on the REACT2023 datasets, and our model ultimately obtains the second place in ACM Multimedia REACT2023 after expert evaluation.

2 RELATED WORK

As discussed in [19], during dyadic interactions, when individuals communicate with each other using both verbal and non-verbal cues, there is a wide range of possible responses (verbal) and reactions (non-verbal) that may be deemed appropriate for an individual (i.e., listener) to generate based on their internal state. The study conducted by Hess et al. [6] revealed that the generation of facial reactions is primarily influenced by individual-specific cognitive processes. These processes are influenced not only by the behavior of the speaker but also by the personality [10] and emotional states [5] of the listener.

Current methods aim to reproduce the Ground Truth (GT) facial reactions of the listener based on the input speaker's behavior. In a previous study, an early approach was introduced that utilized a two-stage conditional generative adversarial network [7]. The objective of this approach was to generate facial reaction sketches by leveraging the input of the speaker's facial action units. Building upon the context of recognizing true personality traits from cognition simulated facial reactions, Song et al. [16, 17] suggested the idea of investigating a specific network for each listener. This proposition enables the modelling of each listener's unique cognitive processes, which in turn, allows for the recreation of their distinct facial reactions, a key factor in identifying true personality traits. Recently, there has been an emergence of studies that delve

into the non-deterministic aspect of the problem, wherein various facial reactions can be predicted from the same input.

3 METHOD

We propose a diffusion based model called FRDiff to tackle the offline multiple appropriate facial reaction generation (OMAFRG) problem. The FRDiff retains the competency of diffusion model to fit distribution, and at the same time can effectively learn from the context. Therefore, our proposed model is better than the simple diffusion models and reaction prediction models. In this section, we first define the problem and notions, and then give a brief introduction of the diffusion model. After that, we describe our model in detail.

3.1 Problem definition

The aim of the OMAFRG task is to train a machine learning model H that takes the entire sequence of speaker behavior $b_{S_n}^{t_1, t_2}$ expressed by a speaker S_n at the time $[t_1; t_2]$ as input and generates a spatiotemporal sequence of multiple appropriate listener facial reactions $P_f(b_{S_n}^{t_1, t_2}) = \{p_f(b_{S_n}^{t_1, t_2})_1, \dots, p_f(b_{S_n}^{t_1, t_2})_M\}$ such that $p_f(b_{S_n}^{t_1, t_2})_1 \neq p_f(b_{S_n}^{t_1, t_2})_2 \neq \dots \neq p_f(b_{S_n}^{t_1, t_2})_M$. Here, M represents the length of the generated sequence. Each generated facial reaction $p_f(b_{S_n}^{t_1, t_2})_i \in P_f(b_{S_n}^{t_1, t_2})$ should be similar to at least one appropriate real facial reaction $F_L(b_{S_n}^{t_1, t_2})$ expressed by human listeners, which can be represented as:

$$F_L(b_{S_n}^{t_1, t_2}) = \begin{bmatrix} f_{L_1}(b_{S_n}^{t_1, t_2})_1 & \dots & f_{L_1}(b_{S_n}^{t_1, t_2})_{I_1} \\ f_{L_2}(b_{S_n}^{t_1, t_2})_1 & \dots & f_{L_2}(b_{S_n}^{t_1, t_2})_{I_2} \\ \dots & \dots & \dots \\ f_{L_N}(b_{S_n}^{t_1, t_2})_1 & \dots & f_{L_N}(b_{S_n}^{t_1, t_2})_{I_N} \end{bmatrix}$$

where $f_{L_\eta}(b_{S_n}^{t_1, t_2})_i$ ($i = 1, 2, \dots, I_\eta$) denotes the appropriate facial reaction that could be expressed by the η_{th} listener in response to the $b_{S_n}^{t_1, t_2}$ under the i_{th} context.

3.2 Diffusion model

Diffusion model is a probabilistic model that aims to learn the original data distribution $P(x)$ by gradually denoising normal distribution variables. This can be interpreted as learning the reverse process of a fixed Markov chain of step length T . In each step, the diffusion model learns to predict the noise added in the forward Markov process using a noise predictor and then removing it. The corresponding objective can be written as:

$$L_{DM} = \mathbb{E}_{x, \epsilon \sim \mathcal{N}(0, 1), t} [\|\epsilon - \epsilon_\theta(x_t, t)\|_2^2] \quad (1)$$

where x_t is a noisy version of the input x and t uniformly sampled from $\{1, \dots, T\}$.

When using diffusion models in high-dimensional data spaces, the curse of dimensionality arises. This is because as the number of features in the data increases, the volume of the high-dimensional space grows much faster than the number of data points, resulting

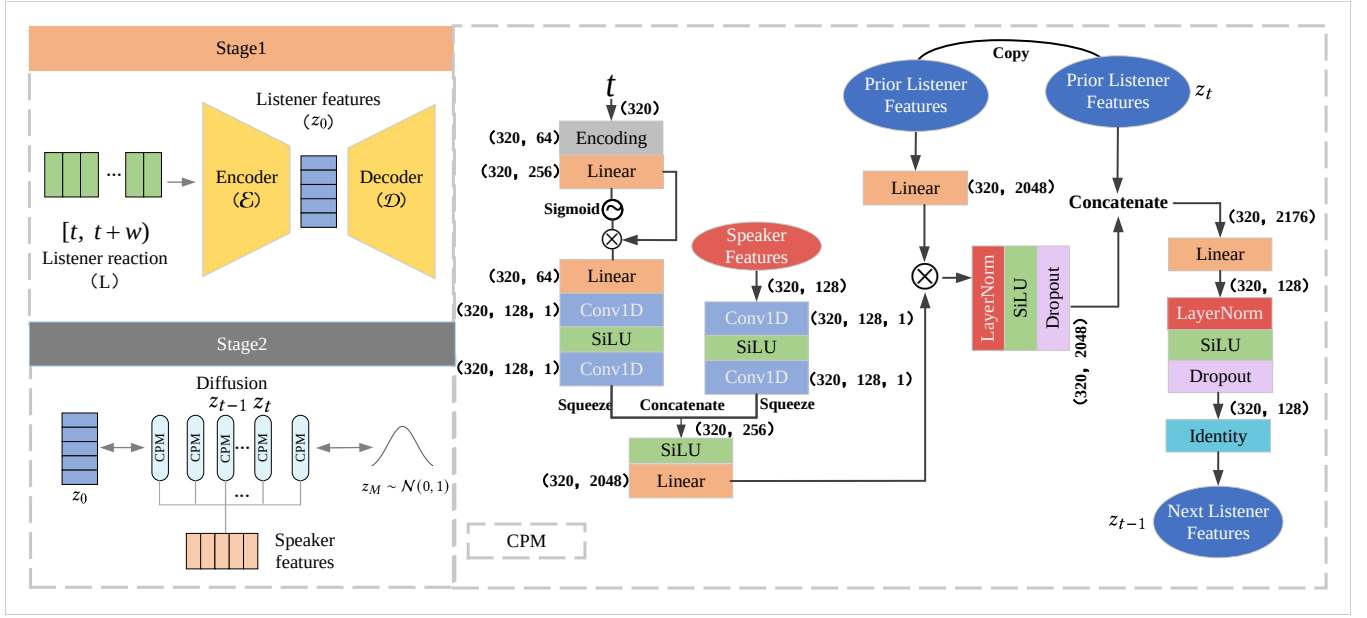


Figure 2: The FRDiff method consists of two stages. In the first stage, the Listener reactions are encoded as latent variable z_0 . In the second stage, a LDM is learned that takes the Speaker features as conditions, with z_0 as the input. During inference, the LDM is used to obtain the latent variable, which is then decoded to reconstruct the Listener reactions.

in very sparse distances between data points. The diffusion models trained on sparse datasets perform badly in learning the data distribution in the high-dimensional space effectively, as it requires too much computation while dealing with sparse data points, which leads to slow diffusion processes and inaccurate results.

The LDM addresses this issue by compressing the original data into an effective, low-dimensional latent space using a trained encoder-decoder model. The objective is simplified as:

$$L_{LDM} := \mathbb{E}_{\mathcal{E}(x), \epsilon \sim \mathcal{N}(0,1), t} [\|\epsilon - \epsilon_\theta(z_t, t)\|_2^2] \quad (2)$$

where z_t is the noisy version of the encoded latent variable z .

3.3 FRDiff model

As is shown in Fig.2, our proposed FRDiff model consists of two parts. And the two parts are trained in two stages separately. In the first stage, we train a variational autoencoders (VAE) [9] based on Recurrent Neural Network (RNN) [21] to convert Listener facial expressions L into a low-dimensional latent space z_0 using the encoder $\mathcal{E}, z_0 = \mathcal{E}(L)$, and the decoder $\mathcal{D}$ can convert z_0 back to the original space, i.e., $L = \mathcal{D}(z_0)$. We constrain it using L_1 loss function and $\mathcal{KL}$ divergence.

$$L_{stage1} = w_1 * L_1 + w_2 * L_2 \quad (3)$$

$$L_1 = -\frac{1}{2} \left(\log \sigma^2 - \sigma^2 - \mu^2 + 1 \right) \quad (4)$$

$$L_2 = \text{MSE}(L, \hat{L}) \quad (5)$$

In the second stage, we first encode Listener facial expressions using the encoder from the first stage, obtaining $z_0 = \mathcal{E}(L)$. Then, we train an LDM conditioned on Speaker facial expression features c to predict the corresponding latent variable z_0 . The process is

explained as follows: the latent variable z_0 undergoes a forward process of a Markov chain with M steps, transforming into $z_M \sim \mathcal{N}(0, 1)$. During training, the network predicts $z_{t-1} = f(z_t, t, c)$ by reversing the Markov chain of length M and using a conditional noise predictor $f(\cdot)$, i.e., CPM (see Fig.2). Different from Belfusion's conditional noise predictor designed by MLP, the CPM of FRDiff is improved by using the Linear layers and Conv1D. We view the generation of facial expressions as events related to only a few past frames. Using 1D convolution can extract local features, which capture local patterns in the input data without considering the relationships between all input features. This process is denoted as $q(z_{t-1} | z_0)$. Therefore, the objective function of the second stage is:

$$\mathcal{L}_{stage2}(c, L) = \sum_{t=1}^T \mathbb{E}_{q(z_t | z_0)} \|f(z_t, t, c) - \mathcal{E}\|_1 \quad (6)$$

During the inference process, we randomly sample z from the $\mathcal{N}(0, 1)$ distribution using c as a condition and reconstruct Listener facial expressions using the decoder $\mathcal{D}$.

4 EXPERIMENTS AND RESULTS

4.1 Experiment settings

Our proposed model is implemented with the Pytorch framework and executed on a platform equipped with 8×3090 GPUs. The training procedure of our model is divided into two stages. In the first stage, the number of epochs for the VAE is set at 1000, the window size is set to 50, and the learning rate is set at 1e-3. In the second stage, the LDM is designated to 100 epochs, the window size is set to 50, the learning rate is adjusted to 1e-4, and the weight decay is set at 5e-4. Given that we utilize only the Emotion and 3D_FV_files,

Table 1: Offline facial reaction generation results achieved on the validation set.

Method	FRC ($\uparrow$)	FRD ($\downarrow$)	FRDiv ($\uparrow$)	FRVar ($\uparrow$)	FRDvs ($\uparrow$)	FRSyn ($\downarrow$)
GT	8.42	0.00	0.0000	0.0666	0.2251	48.52
B_Random	0.04	229.37	0.1667	0.0833	0.1667	46.65
B_Mime	0.35	91.37	0.0000	0.0666	0.2251	44.53
B_MeanSeq	0.01	102.17	0.0000	0.0000	0.0000	47.19
B_MeanFr	0.00	102.84	0.0000	0.0000	0.0000	49.00
Trans-VAE	0.12	101.37	0.0265	0.0045	0.0322	49.00
BeLFusion (k=10)	0.14	94.26	0.0134	0.0078	0.0149	46.94
Ours	0.13	152.51	0.1355	0.0776	0.1356	48.99

Table 2: Offline facial reaction generation results achieved on the test set.

Method	FRC ($\uparrow$)	FRD ($\downarrow$)	FRDiv ($\uparrow$)	FRVar ($\uparrow$)	FRDvs ($\uparrow$)	FRSyn ($\downarrow$)
GT	8.74	0.00	0.0000	0.0723	0.2474	47.72
B_Random	0.04	237.62	0.1667	0.0833	0.1667	43.99
B_Mime	0.38	92.95	0.0000	0.0723	0.2474	38.66
B_MeanSeq	0.01	98.39	0.0000	0.0000	0.0000	45.39
B_MeanFr	0.00	99.04	0.0000	0.0000	0.0000	49.00
Trans-VAE	0.10	98.48	0.0242	0.0040	0.0263	44.88
BeLFusion (k=10)	0.13	89.84	0.0137	0.0078	0.0149	45.02
Ours	0.12	150.53	0.1333	0.0767	0.1335	48.99

the entire training process requires approximately six hours. We conduct experiments on three datasets provided by REACT2023, namely UDIVA [13], RECOLA [14], and NOXI [4]. Moreover, all the data undergoes a mirroring process.

4.2 Results and Analysis

Our proposed method has been evaluated relative to four naive baselines, namely B_Random, B_Mime, B_MeanSeq, and B_MeanFr, as well as two sophisticated baselines, trans-VAE and BeLFusion. The results attained by these methods on the validation and test sets are displayed in Table 1 and 2, respectively. The information conveyed in Table 1 and 2 is largely consistent. Data derived from existing studies [18] are listed in all rows, except for the row labeled 'Ours', where the results achieved by our proposed method are presented. Among all the naive baselines, B_Random, which signifies randomly sampled facial reactions, exhibits admirable diversity, yet proves inappropriate with respect to the FRC and FRD metrics. In contrast, the deterministic models (B_Mime, B_MeanSeq, and B_MeanFr) exhibit superior performance in terms of appropriateness, but lack diversity. The deep learning-based probabilistic models, trans-VAE and BeLFusion, aim to optimally balance appropriateness and diversity. It can be seen from both Table 1 and 2 that these models have not only yielded satisfactory results in terms of appropriateness but have also preserved a degree of diversity.

As is illustrated in Table 1 and 2, both baseline models demonstrate the capacity to generate facial reactions that are in positive correlation with the corresponding actual reactions. Nevertheless, the performance of both the trans-VAE and BeLFusion models leave much to be desired in terms of the diversity metrics. In stark contrast, our method, while possibly not weighting these two models

most appropriately, is still distinctly excels them on the three diversity metrics: FRDiv, FRVar, and FRDvs. Compared with global modeling, local context modeling has higher flexibility, thus brings about more diversity. Indeed, our diversity scores nearly the method B_Random, yet in terms of appropriateness, our method clearly outperforms them.

5 CONCLUSION

In this paper, we propose a new model named FRDiff to tackle the Offline Multiple Appropriate Facial Reaction Generation (OMAFRG) task. The FRDiff model is designed based on the latent diffusion model with modification to enhance the competency of modeling the context. Moreover, our proposed model inherits the stochasticity of the diffusion models, which helps our proposed model to generate multiple reactions. We conduct experiments on the datasets provided by the ACM Multimedia REACT2023 and the results verify that our model satisfies the requirement of randomness and context modeling well. Finally, we obtain the second place in the REACT2023 challenge.

ACKNOWLEDGMENTS

This work was supported by the Natural Science Foundation of China (62276242), National Aviation Science Foundation (2022Z0710 78001), CAAI-Huawei MindSpore Open Fund (CAAIXSJLJJ-2021-016B, CAAIXSJLJJ-2022-001A), Anhui Province Key Research and Development Program (202104a05020007), USTC-IAT Application Sci. & Tech. Achievement Cultivation Program (JL06521001Y), Sci. & Tech. Innovation Special Zone (20-163-14-LZ-001-004-01).

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